

Project: Mean Variance Portfolio Optimizer

In this project, I built a portfolio optimization system to study how risk, return, and data assumptions interact in real financial models.

As I was reading a book on hedge fund investment strategies, one thing I found very interesting was the fact that they use mathematical equations to calculate risk and find an “optimal” portfolio (highest return for the risk taken). In this project, I tried to understand and explore how these portfolios are actually structured.

I began by starting with common financial formulas, and solving them to reach a simplified version that I could use. This way, I tried to understand the mathematics behind the model I was building.

array of individual return data $r_i \approx \log\left(\frac{P_i}{P_{i-1}}\right)$ where P_i is individual prices $r_i \in \mathbb{R}^n$

$$\mu = \mathbb{E}[r_i] \quad \mu \in \mathbb{R}^n$$

$$\Sigma_{xy} = \mathbb{E}[(r_x - \mu_x)(r_y - \mu_y)]$$

$$\Sigma = \begin{bmatrix} (r_1 - \mu_1)(r_1 - \mu_1) & \dots & (r_1 - \mu_1)(r_n - \mu_n) \\ \vdots & & \vdots \\ (r_n - \mu_n)(r_1 - \mu_1) & \dots & (r_n - \mu_n)(r_n - \mu_n) \end{bmatrix} = \mathbb{E}[(r - \mu)(r - \mu)^T]$$

$$\omega \in \mathbb{R}^n \quad r_p = \omega^T r_i$$

$$\mathbb{E}[r_p] = \mathbb{E}[\omega^T r_i] = \omega^T \mathbb{E}[r_i] = \omega^T \mu$$

$$\text{Portfolio variance } V(r_p) = \mathbb{E}[(r_p - \mathbb{E}[r_p])^2]$$

$$= \mathbb{E}[(\omega^T r_i - \omega^T \mu)^2] = \mathbb{E}[\underbrace{(\omega^T (r_i - \mu))^2}_{\text{scalar square}}] = \mathbb{E}[\omega^T (r_i - \mu)(r_i - \mu)^T \omega]$$

$$= \omega^T \mathbb{E}[(r_i - \mu)(r_i - \mu)^T] \omega = \omega^T \Sigma \omega$$

Efficient Frontier

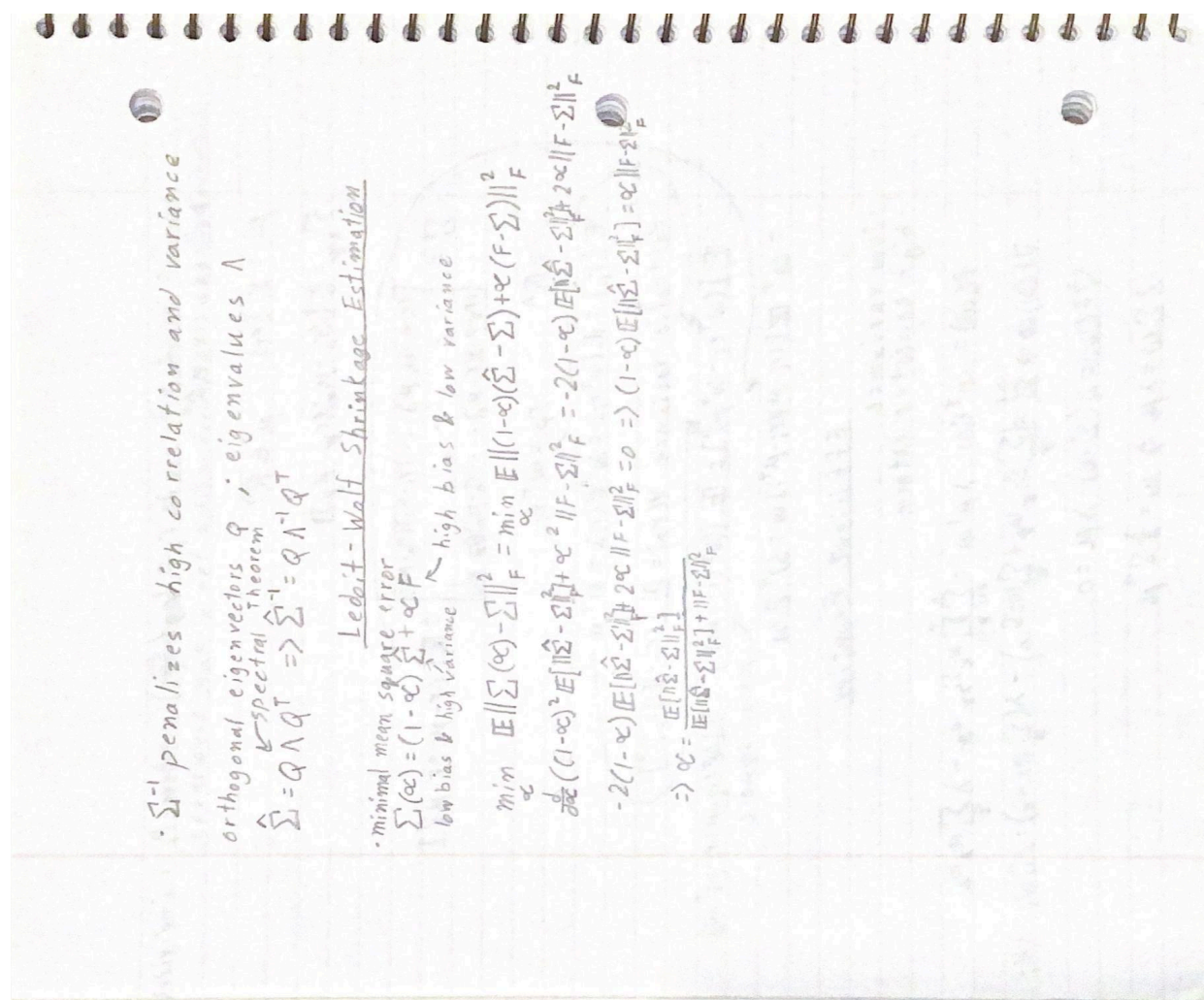
- low variance
- high expected return

$$f(\omega) = \omega^T \Sigma \omega - \lambda \mu^T \omega = \sum_{i=1}^n \omega_i \Sigma_{xy} \omega_y - \lambda \sum_z \mu_z \omega_z$$

$$\nabla f(\omega) = 0 \Rightarrow \frac{\partial f}{\partial \omega_i} = \left(\sum_y \Sigma_{iy} \omega_y + \sum_x \omega_i \Sigma_{ix} \right) - \lambda \left(\sum_z \mu_z \omega_z \right) = 2 \Sigma \omega_i - \lambda \mu_i$$

$$\nabla f(\omega) = 2(\Sigma \omega) - \lambda \mu = 0$$

$$2 \Sigma \omega = \lambda \mu \Rightarrow \omega = \frac{\lambda}{2} \Sigma^{-1} \mu$$



Here, I started with a spectral theorem and the ledoit-wolf shrinkage theorem.

Step 1: Turning Prices Into Mathematical Objects

I start by pulling historical stock price data using Yahoo Finance. Instead of working directly with prices, I convert them into log returns, which is standard in quantitative finance and helps convert changing prices into a standard scale (return during that time period).

From these returns, I compute an expected return (the average return of each asset) and a covariance matrix, which measures how assets move together and determines portfolio risk.

Step 2: Stabilizing the Risk Model (Shrinkage)

Raw covariance matrices estimated from data are often noisy, especially with limited samples. Small errors can lead to extreme and unrealistic portfolio weights.

To address this, I apply covariance shrinkage, blending the sample covariance matrix with a diagonal matrix that assumes assets are uncorrelated. This reduces overfitting while still preserving meaningful structure in the data.

This is a technique used in professional quantitative finance.

Step 3: Solving the Optimization Numerically

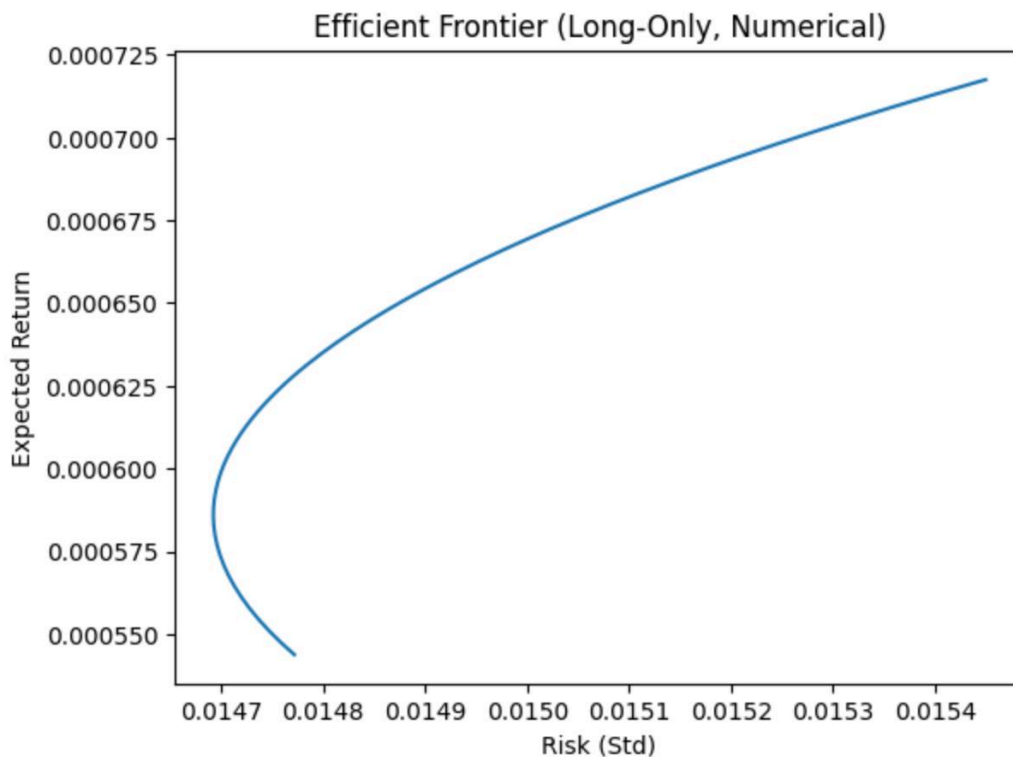
Rather than using a closed-form solution, I implement gradient descent to solve the optimization problem numerically.

The objective function balances risk (portfolio variance), return (expected portfolio return), and a changeable risk-aversion parameter(λ)

After each gradient step, I project the solution back onto a feasible set where all weights are non-negative (only long positions a.k.a only buying stocks, not shorting them) and all weights sum to 1 (to find percentage allocations)

Step 4: Tracing the Efficient Frontier

By varying the risk-aversion parameter, I generate an entire family of portfolios and plot the efficient frontier, which shows the best possible expected return for each level of risk.



From this frontier, I found the minimum-variance portfolio(lowest risk) and the maximum Sharpe ratio (tangency) portfolio, using real risk-free rate(current borrowing rate). The risk-free rate is useful because if someone wants to apply margin to the portfolio(borrow money to increase leverage), the risk-free rate being the starting point is essential. This also helps compare if it's better to buy this portfolio or to just get government bonds.

Minimum Variance Portfolio:
Expected Return: 0.010839
Volatility: 0.030850

Out[]: **Asset Weights**

Ticker	
AMZN	0.343771
BA	0.286849
INFY	0.369380

What's the current 1 year Treasury Yield Rate?: 0.0345
Max Sharpe Ratio: 0.3382110676903233
Expected Returns: 0.014402672684607098
Volatility: 0.03421578191164793
Portfolio for Max Sharpe Ratio:

Out[]: **Asset Weights**

Ticker	
AMZN	0.610414
BA	0.134751
INFY	0.254836

Step 5: Daily vs. Monthly Data

Initially, I assumed that using daily returns would give better estimates because it provides more data. Instead, I observed extreme sensitivity, small changes in the data window led to large swings in portfolio weights.

This led me to test the same optimization using monthly returns. Monthly data reduced short-term noise caused by earnings announcements, sentiment shifts, and random fluctuations.

The results were much more stable and estimated volatility dropped significantly while the estimated returns stayed relatively the same.

Annualized return using daily data: 0.175817671398244

Annualized variance using daily data: 0.05703144953164485

Annualized volatility(standard deviation) using daily data: 39.90818833499823

Annualized return using monthly data: 0.18698582070686864

Annualized variance using monthly data: 0.014140212112553563

Annualized volatility(standard deviation) using monthly data: 0.4974024856437296

Why This Matters

This work reinforced my interest in understanding systems at a structural level. Rather than treating optimization as a black box, I learned how fragile these models can be, and how careful design choices can make them more robust and meaningful when investing in stocks.